

Medallion Architecture implementation using Databricks

Part 2

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Dataset Explanation

The dataset used for this workshop is titled `sample_employee_dataset.csv`. It is a small, structured dataset designed to simulate real-world employee records and demonstrate end-to-end data processing using the Medallion Architecture on Databricks.

Column	Description	Example
Employee_ID	Unique identifier assigned to each employee	E101
Name	Full name of the employee	Alice Johnson
Department	Business function or division the employee works in	Marketing
Performance_Score	Numeric performance evaluation score (1–100 scale)	88
Monthly_Salary	Employee's monthly pay in USD	5400

Overall Approach with Screenshots

The workshop follows the Medallion Architecture design, a layered data-engineering pattern used in Databricks to incrementally refine data through Raw, Bronze, Silver, and Gold stages.

1. Raw Layer

- Purpose: Stores the CSV data exactly as received without any transformation.
- Implementation: The dataset is uploaded into a Raw table under a schema named `employee_demo`.

2. Bronze Layer

- Purpose: Handles streaming ingestion of files from the landing zone using Databricks Auto Loader.

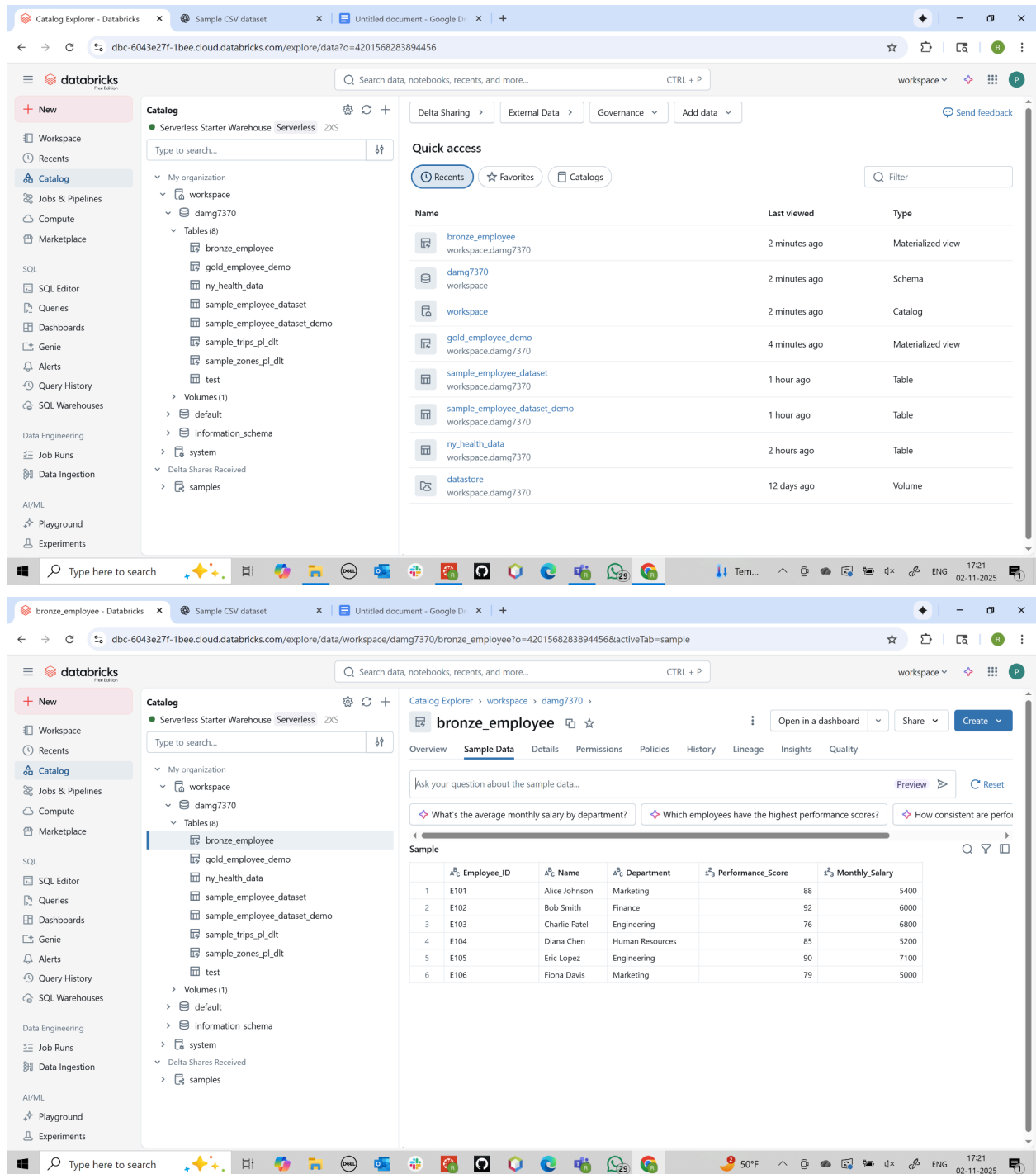
- Key action: Reads CSV files as a continuous stream and appends new data automatically to the bronze_employee table.
- Validation: Confirms ingestion by checking record counts and sample rows.
- Screenshot: Figure 1 shows the pipeline configuration screen with the Auto Loader setup.

The screenshot shows the Databricks workspace interface for a pipeline named 'PL_DLT'. The pipeline is configured with three stages: 'bronze_employee', 'silver_employee', and 'gold_employee'. The 'silver_employee' stage is currently selected, and its configuration is shown in the 'Tables' tab. The table below lists the pipeline's components and their status.

Name	Catalog	Schema	Type	Durati...	Outpu...	Expectations	Drop...	Warni...	Failed	Incrementaliza...
silver_employee	N/A	N/A	View	-	-	Not defined	-	-	-	N/A
bronze_employee	workspace	damg7370	Materialized vi...	4s	6	Not defined	-	-	-	Full recomput
gold_employee_de...	workspace	damg7370	Materialized vi...	4s	6	Not defined	-	-	-	Full recomput
sample_trips_pl_dlt	workspace	damg7370	Materialized vi...	3s	22K	Not defined	-	-	-	No change
sample_zones_pl_dlt	workspace	damg7370	Materialized vi...	2s	128	Not defined	-	-	-	No change

3. Silver Layer

- Purpose: Cleans and standardizes data types.
- Transformations performed:
 - Trimmed whitespace in text fields (Employee_ID, Name, Department).
 - Cast Performance_Score to integer and Monthly_Salary to double.
- Output: A consistent and analysis-ready table named silver_employee.
- Screenshot: Figure 2 shows the successful DLT pipeline run where Bronze-Silver-Gold flow completed without errors.
- Screenshot: Figure 3 displays the Bronze table preview verifying raw data ingestion.



4. Gold Layer

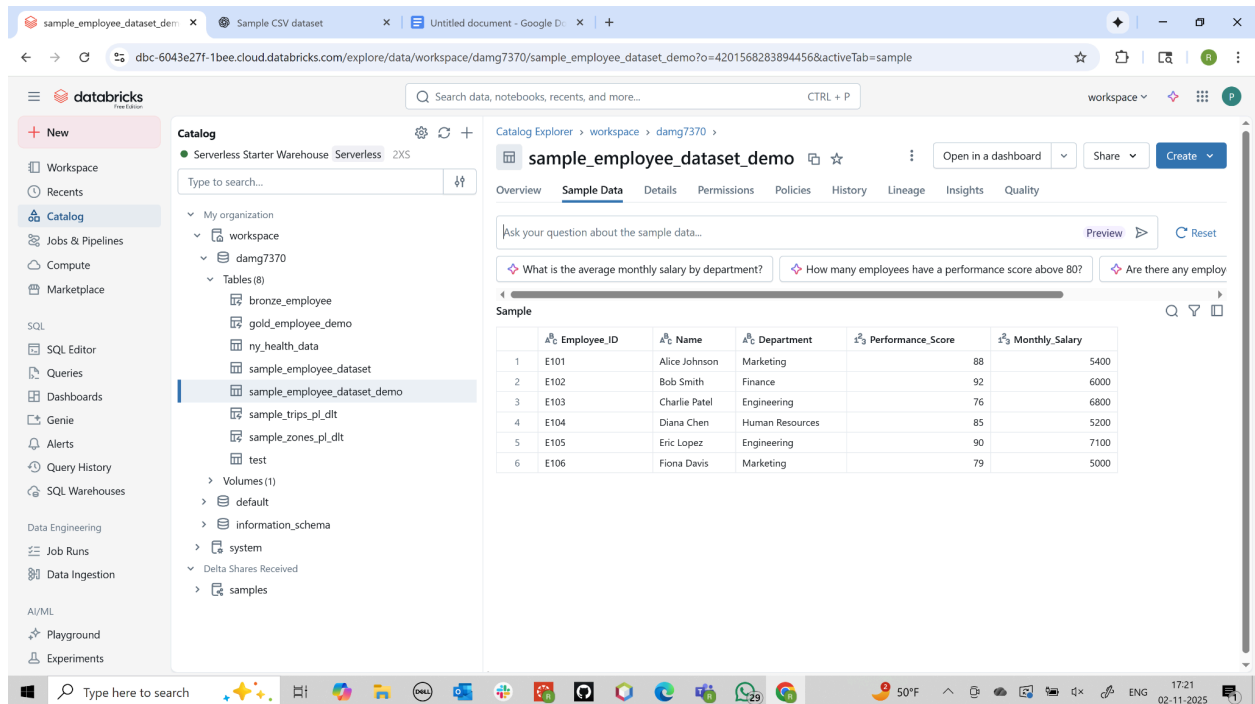
- Purpose: Produces business-level summaries that are consumable for reporting and analytics.
- Output: Table gold_employee_dept_summary used for analytics dashboards.
- Screenshot: Figure 4 illustrates the Gold table view with aggregated data.

The screenshot shows the Databricks Catalog Explorer interface. The left sidebar contains navigation options: New, Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie, Alerts, Query History, SQL Warehouses, Data Engineering, Job Runs, Data Ingestion, AI/ML, Playground, and Experiments. The main panel displays the 'gold_employee_demo' dataset under the 'workspace' catalog. The dataset is shown as a table with the following columns: Employee_ID, Name, Department, Performance_Score, Monthly_Salary, and row_status. The table contains six rows of data, including a new record (E106, 'Fiona Davis', Marketing, 79, 5000.0) that has been added to the dataset.

	Employee_ID	Name	Department	Performance_Score	Monthly_Salary	row_status
1	E101	Alice Johnson	Marketing	88	5400.0	New
2	E102	Bob Smith	Finance	92	6000.0	New
3	E103	Charlie Patel	Engineering	76	6800.0	New
4	E104	Diana Chen	Human Resources	85	5200.0	New
5	E105	Eric Lopez	Engineering	90	7100.0	New
6	E106	Fiona Davis	Marketing	79	5000.0	New

5. Incremental Update Demonstration

- Inserted a new record (E106, “Fiona Davis”, Marketing, 79, 5000) into the Raw table.
- Re-ran the pipeline to confirm the new record propagated automatically through Bronze-Silver-Gold.
- Screenshot: Figure 5 shows the Raw dataset preview, demonstrating that all six original plus the new record appear as expected.



Conclusion

Using sample_employee_dataset.csv, the workshop demonstrates a full medallion pipeline:

- RAW landing
- Bronze streaming ingest
- Gold business aggregation

The post-insert re-run proves incremental data flows through the layers without full rebuilds, aligning with Databricks best practices for reliable, scalable analytics.